Assignment-4 Coding Exercises

Due date: April 19, 2026 – Midnight

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What to submit:

A link to Google Colab or GitHub for Juypter notebook. Feel free to use coding assistants such as GitHub Co-pilot. Anything handed in after the due date will be penalized by 50% for each 24 hours of lateness

Instructions:

Provide a well-organized and properly commented Jupyter Notebook presenting a comparative study of at least two heuristic methods to solve ONLY ONE of the 5 provided problems (operator scheduling, trip itinerary planning, gear train design, ridesharing, and step-cone pulley). The focus of this assignment is on implementation quality, clarity of code, and depth of analysis.

To evaluate the effectiveness of the proposed heuristic algorithms, students are expected to conduct a comprehensive analysis of the results. This may include, but is not limited to, the following:

- Statistical performance indicators, such as Minimum (Min), Average (Ave.), Median (Med), and Standard Deviation (Std)
- Computational performance analysis, to evaluate efficiency in terms of execution time (or time per iteration/generation) and the number of iterations/generations required for convergence.
- Non-parametric statistical tests, such as the Wilcoxon Rank-Sum Test (WRT) and Friedman Ranking Test (FRT), to assess the statistical significance of performance differences
- Convergence analysis, to examine the optimization behavior over iterations
- Scalability analysis, to evaluate performance as the problem size increases
- Sensitivity analysis, to study the effect of key parameters on algorithm performance
- Ablation study, to assess the contribution of individual components or design choices within the heuristic algorithms. For example, in PSO, this can be done by removing or modifying components such as inertia weight, cognitive term, or social term, and analyzing their impact on performance and convergence.

Students are encouraged to incorporate additional relevant evaluation methods or visualizations where appropriate. Clear interpretation, critical discussion of results, and well-structured presentation are essential components of the assignment.

Rubric

Criterion	Description	Marks
Implementation Quality	Correct and efficient implementation of at least two heuristic algorithms	1
Code Quality & Readability	Clean, well-structured, and properly commented code	1
Experimental Setup	Fair and well-designed experiments (parameters, number of runs, and comparison setup)	1
Statistical Analysis	Proper computation and interpretation of Min, Ave., Std, WRT, and FRT, along with thorough analysis of convergence, scalability, sensitivity, and clarity of discussion	2
Bonus	Comparison against exact methods, automated adaptation and tuning strategies, or hybrid methods (well-justified & evaluated)	1

Problems [Select ONLY ONE]

1. A **manufacturing plant** wants to optimize the scheduling of its machine operators to minimize overall labor costs while ensuring that enough workers are available to meet production demand. Each operator has a different hourly wage, and production demand varies throughout the day. The plant must schedule at least **4 operators** during peak production hours and at least **2 operators** during non-peak hours.

The plant can hire **part-time operators** at a lower hourly rate, but they are only available during non-peak hours. The plant also has the option of paying **overtime to full-time operators** to meet demand during peak hours, but at a higher hourly rate.

The goal is to find the **optimal workforce schedule** that meets demand while minimizing total labor costs.

Mathematical Formulation

Objective Function:

Minimize total labor cost:

$$f(x) = \sum(c_i . x_i) + \sum(c_{io} . x_{io}) + \sum(c_{jp} . x_{jp})$$

Constraints:

- **Peak-hour staffing requirement:**

$\sum x_i + \sum x_{io} \geq 4$ (At least 4 full-time operators must be scheduled during peak hours.)

- **Non-peak-hour staffing requirement:**

$\sum x_{jp} \geq 2$ (At least 2 part-time operators must be scheduled during non-peak hours.)

- **Non-negativity constraints:**

$x_i, x_{io}, x_{jp} \geq 0$

Parameters:

- i = index for full-time operators, i =1,2,..., m
- j = index for part-time operators, j =1,2,...,n
- c_i = hourly wage of full-time operator i
- c_{io} = overtime hourly wage of full-time operator i (1.5× full-time rate)
- c_{jp} = hourly wage of part-time operator j

Decision Variables:

- x_i = number of hours worked by full-time operator i during peak hours
- x_{io} = number of overtime hours worked by full-time operator i during peak hours
- x_{jp} = number of hours worked by part-time operator j during non-peak hours

Assume the full-time rates of full-time operators are **[30, 35, 40, 45, 50]**, their overtime rates are **1.5 times** the full-time rates and the part time operators' rates are **[25, 27, 29, 31, 33]**.

2. Consider the **trip itinerary planning problem** where the objective is to provide an optimal travel itinerary for tourists when visiting a new city taking into consideration the quality of the places to be visited, the proximity of the attractions to one another, and finally how much of the day can be completely occupied having little idle time. The problem is stated as following:

Find X which optimizes

$$f(x) = \frac{1}{1 + \frac{Z_c}{480}} \times Z_r \times \frac{Z_d}{480}$$

Subject to the following constraints:

- Duration of itinerary must not exceed total duration in a given day. This is set to 480 minutes in a day. This is expressed as the summation of the duration of each venue and total commute time to be visited in an itinerary. This means that z_d+z_c<=480 minutes
- There should never be repetitions of the same venue in a solution.

where

$X=[z_d\ z_c\ z_r]^T$

z_d represents the duration of the itinerary (minutes).

z_c represents the total commute time in the itinerary.

z_r represents the average rating of all the venues in an itinerary. The ratings are provided from Yelp, Google reviews or others.

Given the following commute times between 10 attractions

Commute Time (mins)		To								
		1	2	3	4	5	6	7	8	9
From	0	10	20	7	11	8	19	7	1	1
	1	0	12	14	1	7	12	10	22	22
	2	12	0	25	21	28	1	22	12	12
	3	14	25	0	15	15	25	4	6	6
	4	1	21	15	0	25	22	11	10	10
	5	7	28	15	25	0	28	14	28	29
	6	12	1	25	22	28	0	22	12	12
	7	10	22	4	11	14	22	0	3	2
	8	22	12	6	10	28	12	3	0	1
	9	22	12	6	10	29	12	2	1	0

The duration and ratings for each attraction are given as follows:

ID	Rating	Duration (mins)
0	Starting hotel	
1	2	120
2	3	60
3	3	180
4	0	180
5	5	120
6	1	60
7	4	60
8	0	60
9	2	120

3. **Gear train design problem** involves determining the number of teeth for each gear wheel (*A, B, C, and D*) as illustrated in Figure 1.

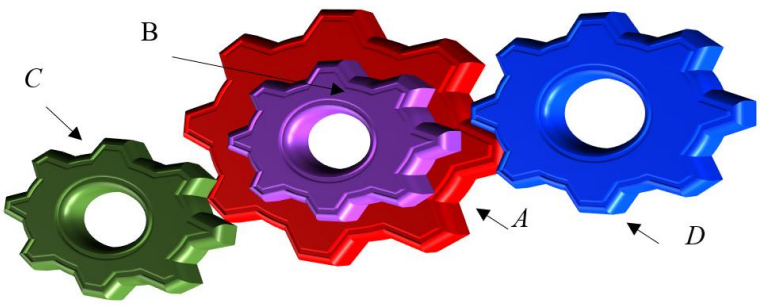

Figure 1 The structure of gear train design problem

The main objective is to minimize the overall gear ratio in the compound gear train layout. The gear train ratio represents the relationship between the angular velocities of the output and input shafts. To achieve the desired ratio, the compound gear train is formed by combining two sets of gear pairs, ensuring the most efficient transmission performance. The mathematical formulation of the total gear ratio is expressed as follows:

$$i_{\text{tot}} = \frac{\omega_o}{\omega_i} = \frac{z_d z_b}{z_a z_f}$$

In this problem, z denotes the total number of teeth on each gearwheel, while ω_I and ω_o represent the angular velocities of the input and output shafts, respectively. The objective is to determine the optimal number of teeth for each gearwheel so that the resulting gear ratio closely matches the target value ($i_{trg} = 1/6.931$). Each gearwheel can have between 12 and 60 teeth. The mathematical formulation of this problem is defined as follows.

$$f(\bar{x}) = \left(i_{trg} - i_{tot}\right)^2 = \left(\frac{1}{6.931} - \frac{x_1 x_2}{x_3 x_4}\right)^2$$

subject to:

$$g_{1-4}(\bar{x}) = 12 - x_i \leq 0$$
$$g_{5-8}(\bar{x}) = (60 - \bar{x}) \leq 0$$

4. **Ridesharing** is a successful implementation of sharing economy business model where personal vehicles are shared by their owners/drivers with individual travelers who have similar itineraries and schedules. Ridesharing problem is a multi-objective constrained optimization problem. A non-comprehensive list of optimization goals for ridesharing includes:

- Minimizing the total travel distance/time of drivers' trips
- Minimizing the total travel time of passengers' trips
- Maximizing the number of matched (served) requests
- Minimizing the cost for the drivers' trips
- Minimizing the cost for the passengers' trips
- Maximizing driver's earning
- Minimizing passenger's waiting time
- Minimizing the total number of drivers required

Consider the ridesharing problem shown in Figure 2 where the objective is to come up with a schedule for drivers that minimizes the total distances of vehicles' trip. The statement of this optimization is as follows:

Find s that minimizes

$$f = \sum_{m \in V} \sum_{i.j \in P} s_{i,j}^{m} d_{i,j}$$

where

f is the total distance of vehicles' trips to serve all the passengers.

$s_{i,j}$ is 1 if location j is assigned within the schedule of the driver and 0 otherwise

$d_{i,j}$ is the distance between two points within the route where $d_{i,j} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$.

P is the passenger set with known pickup and delivery points

V is the vehicle set with predefined initial and final locations

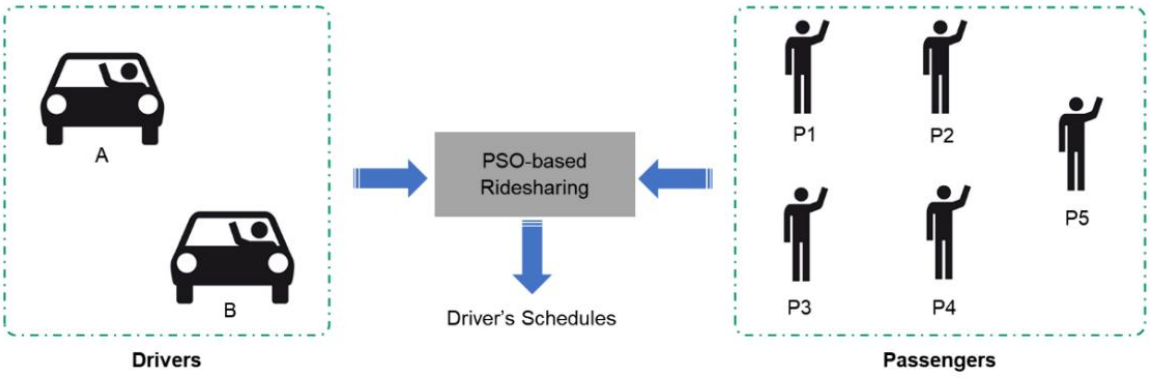

Figure 2 Ridesharing Problem

Given the following pickup and drop off location of 5 passengers

Passengers	Pickup x-coordinate	Pickup y-coordinate	Drop off x-coordinate	Drop off y-coordinate
P1	9	9	4	2
P2	8	5	2	4
P3	6	1	8	6
P4	7	6	9	9

and initial and final locations of the two drivers are

Drivers	Initial x-coordinate	Initial y-coordinate	Final x-coordinate	Final-y coordinate
A	4	1	8	10
B	1	6	9	4

5. **Step-cone pulley problem** aims to minimize the weight of a four-step cone pulley using five design variables. Among these variables, four represent the diameters of the pulley steps (d_1, d_2, d_3, d_4), while the fifth corresponds to the pulley width (ω).

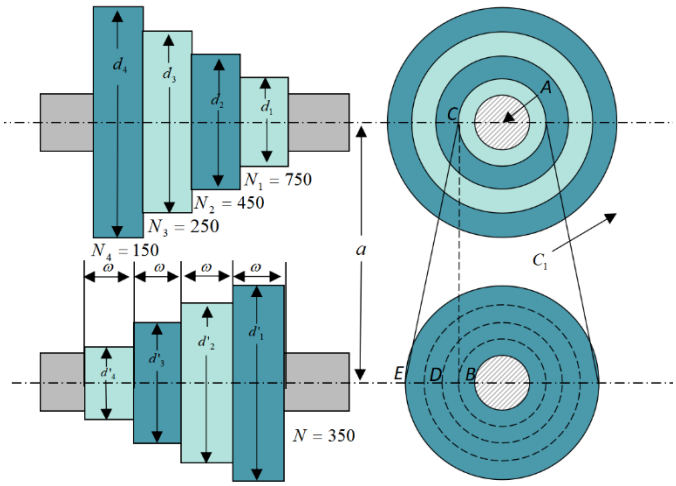

Figure 3 The structure of step-cone pulley problem

The problem involves eleven nonlinear constraints that ensure the transmitted power remains at 0.75 hp. Mathematically, this problem can be defined as follows.

$$f(\bar{x}) = \rho\omega\left[d_1^2\left\{11 + \left(\frac{N_1}{N}\right)^2\right\} + d_2^2\left\{1 + \left(\frac{N_2}{N}\right)^2\right\} + d_3^2\left\{1 + \left(\frac{N_3}{N}\right)^2\right\} + d_4^2\left\{1 + \left(\frac{N_4}{N}\right)^2\right\}\right]$$

subject to:

$$\begin{aligned}
&h_1(\bar{x}) = C_1 - C_2 = 0,\\
&h_2(\bar{x}) = C_1 - C_3 = 0,\\
&h_3(\bar{x}) = C_1 - C_4 = 0,\\
&g_{i=1,2,3,4}(\bar{x}) = -R_i \le 2,\\
&g_{i=5,6,7,8}(\bar{x}) = (0.75 \times 745.6998) - P_i \le 0
\end{aligned}$$

where,

$$\begin{aligned}
&C_i = \frac{\pi d_i}{2}\left(1 + \frac{N_i}{N}\right) + \frac{\left(\frac{N_i}{N} - 1\right)^2}{4a} + 2a, i = (1,2,3,4),\\
&R_i = \exp\left(\mu\left\{\pi - 2\sin^{-1}\left\{\left(\frac{N_i}{N} - 1\right)\frac{d_i}{2a}\right\}\right\}\right), i = (1,2,3,4)\\
&P_i = \mathrm{st}\omega(1 - R_i)\frac{\pi d_i N_i}{60}, i = (1,2,3,4)\\
&t = 8\ \mathrm{mm}, s = 1.75\mathrm{MPa}, \mu = 0.35, \rho = 7200\ \mathrm{kg/m^3}, a = 3\ \mathrm{mm}
\end{aligned}$$